Ibn Sina National College for Medical Studies

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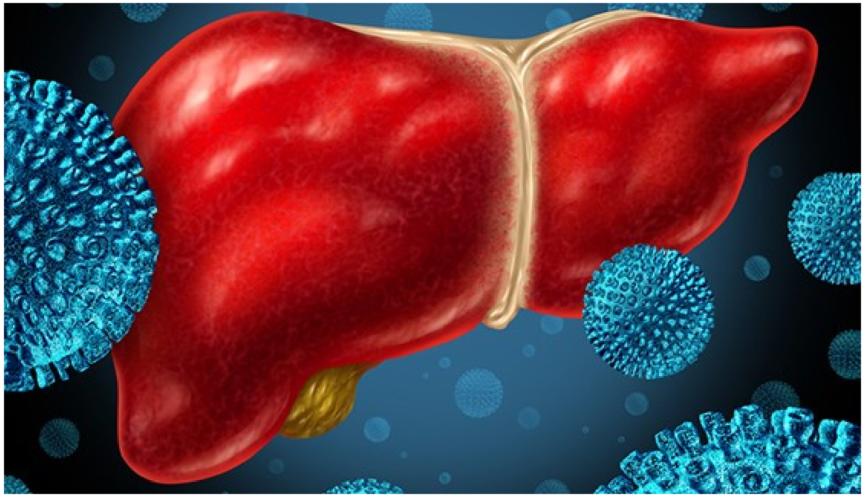
HEPATITIS



Rehab A Omran

Prof Of Internal Medicine

Lecture Learning Outcomes

By the end of this lesson, students should be able to:

1. Identify the types of hepatitis relevant to dental practice, including hepatitis A, B, C, D, and E.
2. Describe the transmission routes of hepatitis viruses in the dental setting, emphasizing bloodborne transmission and potential exposure pathways.
3. Recognize the signs and symptoms of hepatitis infection, particularly those relevant to dental professionals
4. Understand the importance of infection control measures in preventing the transmission of hepatitis in dental practice, including hand hygiene, personal protective equipment (PPE), and instrument sterilization protocols.
5. Demonstrate the correct use of barrier techniques, such as gloves, masks, and protective eyewear, to minimize the risk of hepatitis transmission during dental procedures.
6. Explain the role of immunization in preventing hepatitis infection among dental healthcare workers and patients, including the recommendation for hepatitis B vaccination and potential considerations for other hepatitis vaccines.

Hepatitis

- **ACUTE HEPATITIS:** Acute parenchymal liver damage can be caused by many agents as viruses, toxins and drugs.

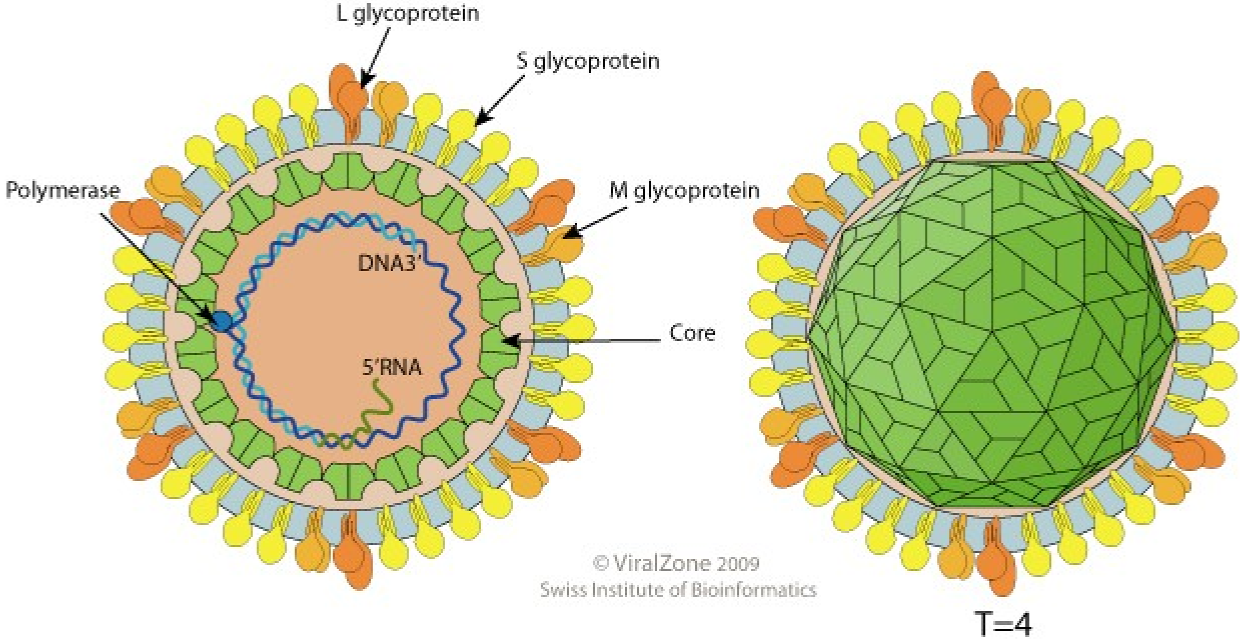
- **CHRONIC HEPATITIS:** Any hepatitis lasting for 6 months or longer and is classified according to the etiology.

- Chronic viral disease is the principal cause of chronic liver disease, cirrhosis and hepatocellular carcinoma worldwide.

Characteristics Of Viral Hepatitis

	HAV	HBV	HCV	HDV	HEV
Genome	RNA	DNA	RNA	RNA	RNA
Family	Picorna	Hepadna	Flavi	viroid	
IP (days)	15-45	30-180	15-150	30-80	15-60
transmission	Feco-oral	Blood Saliva	Blood Saliva	Blood	Feco-oral
Serologic diagnosis	Ig M anti-HAV	Ig M anti-HBV	Anti-HCV HCV RNA	Ig M anti-HDV	Ig M anti-HEV
Carrier state	No	Yes	?	Yes	No
Chronicity	-	+	+	+	-
Prevention	Vaccine	vaccine		Vaccination against HBV	

HBV


L glycoprotein
S glycoprotein
Polymerase
M glycoprotein
DNA3'
Core
5'RNA
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T=4

HEPATITIS B [HBV]

Epidemiology:

- The hepatitis B is present worldwide with an estimated 360 million carriers

Transmission:

- Vertical transmission from mother to child
- Horizontal transmission by intravenous route or by close personal contact. The virus can be found in semen and saliva

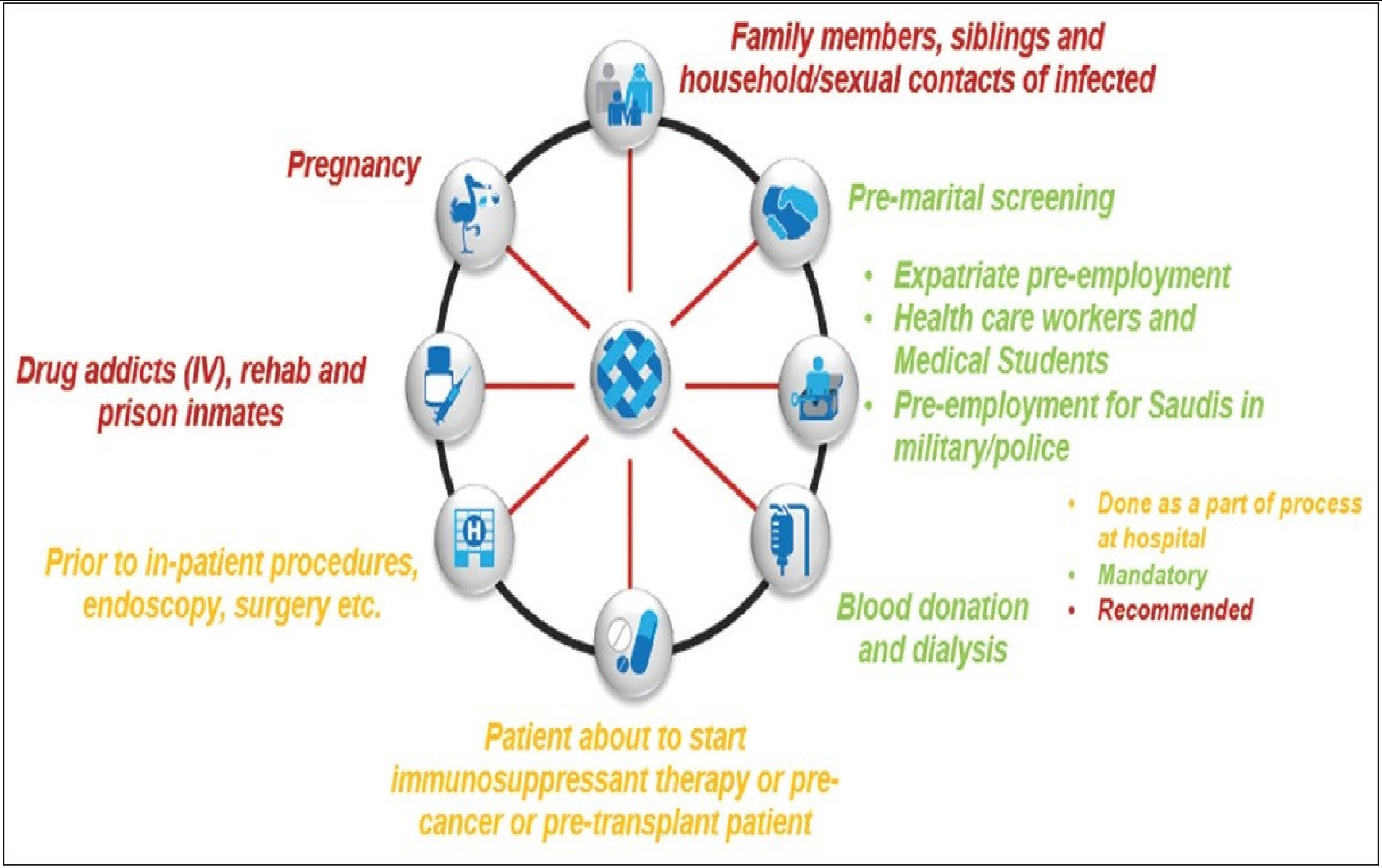
Hepatitis B

- Hepatitis B, a major cause of liver damage and liver cancer, is a silent epidemic worldwide. **It is a blood borne infection that is commonly transmitted through blood sexual fluids and vertically.**

- **In the dental office**, hepatitis B is mostly **transmitted via** saliva ,needle prick injuries and blood of carriers . HBV spreads through oral secretions and gingival crevicular fluid have been noticed

- unvaccinated dental **health care workers have a 10 times greater risk** of becoming infected with hepatitis B because of possible occupational exposure

Potential HBV screening points in KSA



HBV serological markers

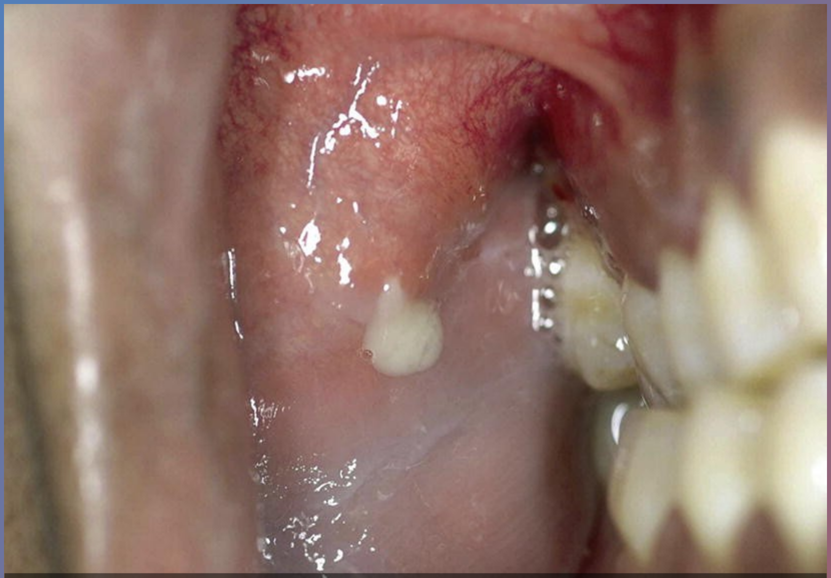
Test	Clinical interpretation
HBsAg (hepatitis B surface antigen)	**Hallmark of infection** Positive in the early phase of acute infection and persists in chronic infection Quantification of HBsAg is a potential alternative marker of viraemia and it is also used to monitor the response to antiviral treatment
Anti-HBc IgM (hepatitis B core antibody)	IgM subclass of anti-HBc and observed during **acute infection** (used to differentiate between acute and chronic HBV infection) Might become positive during severe exacerbation of chronic infection
Anti-HBc (total)	Develops around 3 months after infection (most constant marker of infection) Total anti-HBc (IgM, IgA and IgG) indicates **resolved infection**
HBeAg (hepatitis B e antigen)	Viral protein usually associated with **high viral load** and **high infectivity**
Anti-HBe (hepatitis B e antibody)	Antibody to HBeAg usually indicates decreasing HBV DNA But present in the immune-control and immune-escape phases
Anti-HBs (hepatitis B surface antibody)	Neutralizing antibody that confers protection from infection Recovery from acute infection (with anti-HBc IgG) Immunity from vaccination

Interpretation of hepatitis B serologic test results

Clinical State	HBsAg	Anti-HBs	Total Anti-HBc	Action
Acute infection	Positive	Negative	Positive (IgM positive)	Link to hepatitis B care
Chronic infection	Positive	Negative	Positive (IgM negative)	Link to hepatitis B care
Resolved infection	Negative	Positive	Positive	Counsel
Immune from vaccination	Negative	Positive	Negative	Reassure if history of HepB vaccine series completion
Susceptible, never infected	Negative	Negative	Negative	Offer HepB vaccine if no history of HepB vaccine series completion
Isolated core antibody positive	Negative	Negative	Positive	Consult with specialist

Hepatitis C

- Less widespread, rarely transmitted during dentistry.
- **Less readily transmitted by needle stick injuries.**
- More vulnerable to antiseptics.
- **Acute hepatitis is uncommon, usually mild. But:**
- **High risk of HCC (10 %).**
- **Infection persists in 80%.**
- **No protective vaccine.**
- More common in IV drug addicts, tattooing.
- Most HCV patients had higher HCV RNA levels in their gingival sulcus than in their saliva.
- HCV-RNA could be found in tooth brushes used by hepatitis C patients



Sialadenitis

lichen planus

Oral manifestations of hepatitis B and C infection

lichen planus

Sjögrens syndrome

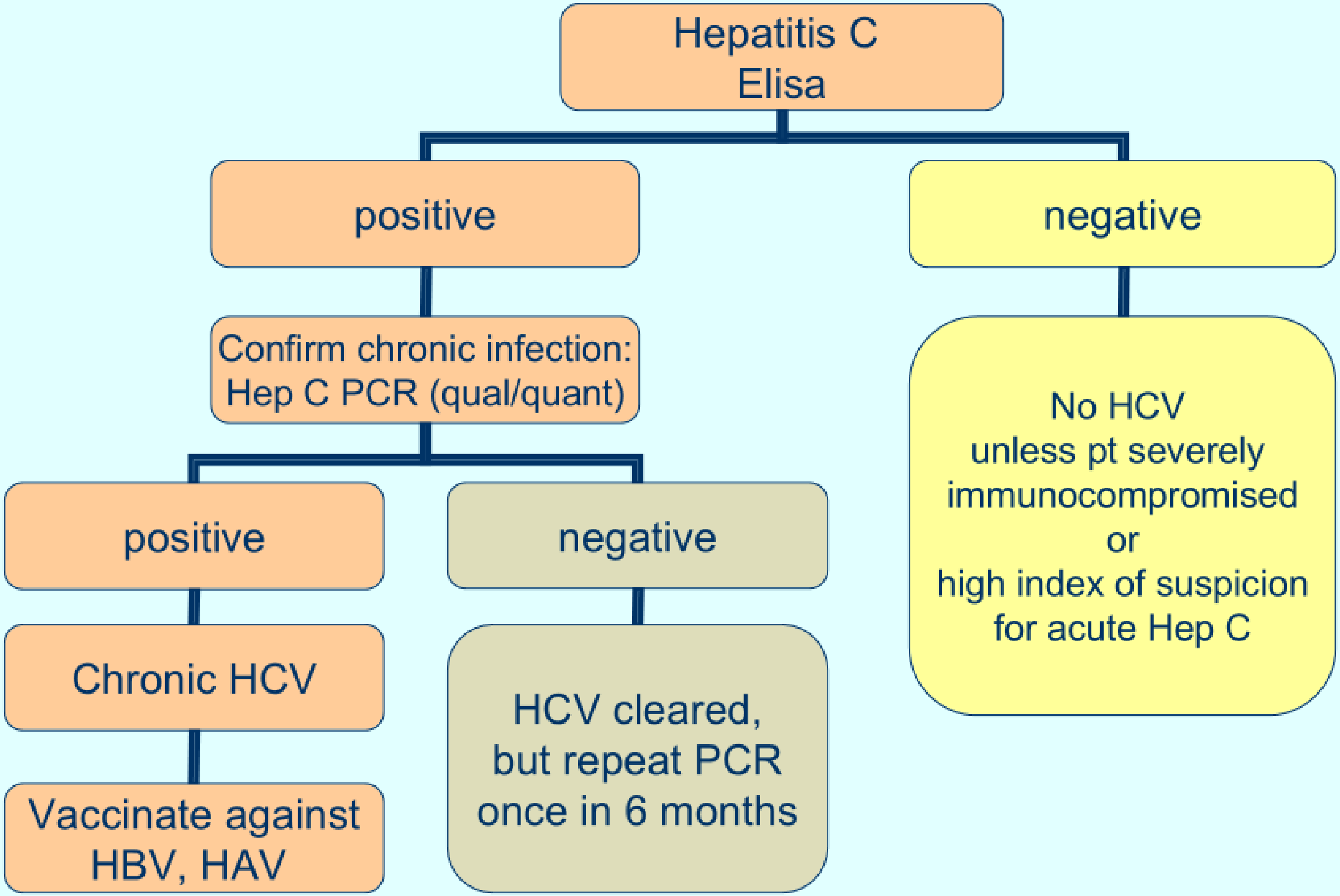
Sialadenitis

Some forms of oral cancers may also be seen.

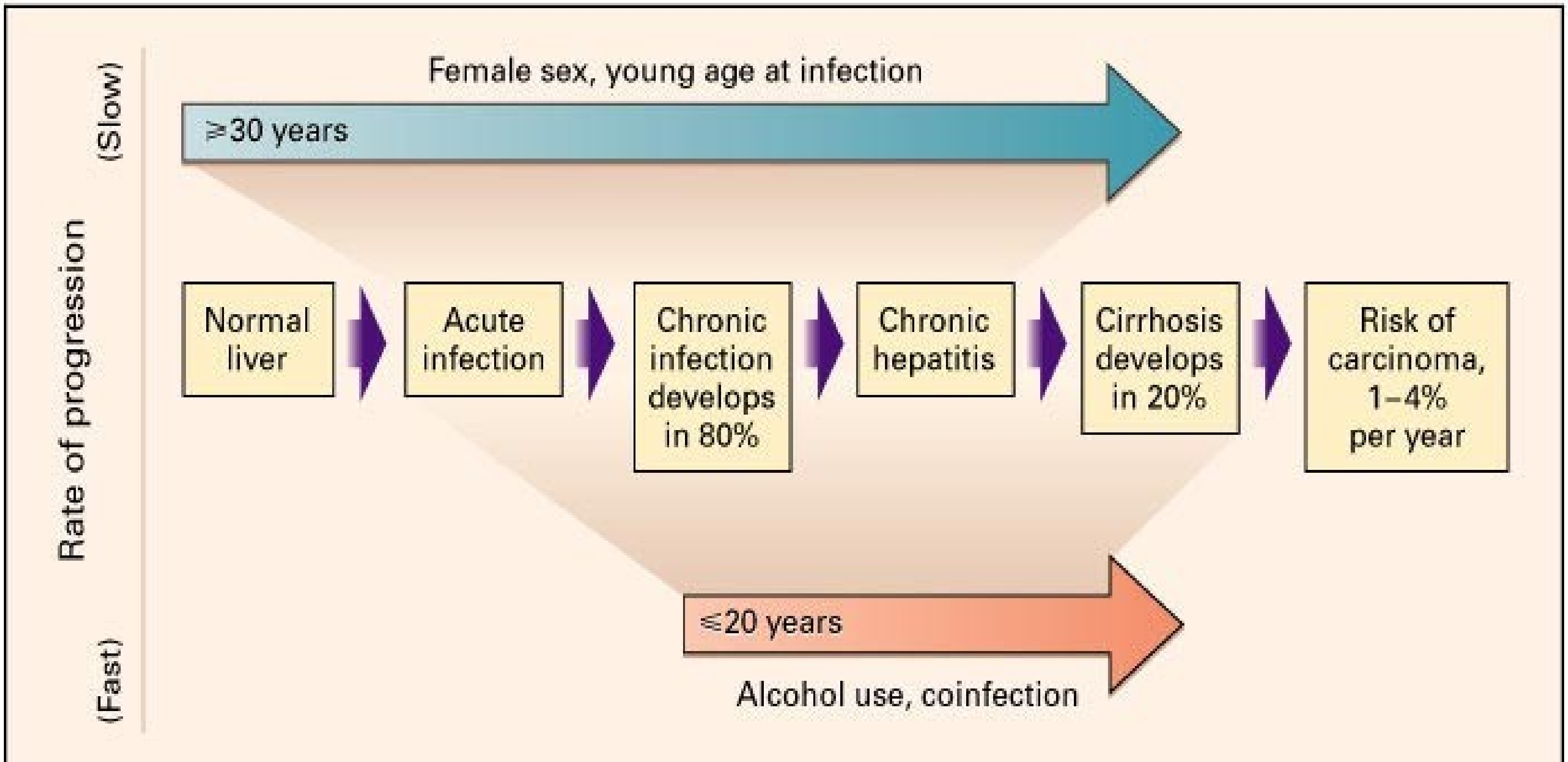
Cirrhotic patients may have thrombocytopenia due to hypersplenism or treatment with interferon.

In patients with liver disease, the resultant impaired hemostasis can be manifested as petechiae or excessive gingival bleeding with minor trauma. **This is especially suggestive if it occurs in the absence of inflammation**.

Diagnostic approach to hepatitis C virus infection



Natural Course of Hepatitis C



PROGNOSIS

- 80% of acute hepatitis C become chronic (of these 20% evolve to cirrhosis)
- Risk of hepatocellular carcinoma increases if cirrhotic
- Can cause cryoglobulinemia; associated with membranoproliferative glomerulonephritis, lymphoma

How to prevent

- All health care providers, including those working in dentistry should be **vaccinated** to protect them.
- **Patients** can also play a role by ensuring that they are **vaccinated as well**. The hepatitis B vaccine is safe and effective and protects for a lifetime!

- **If a patient with active hepatitis, positive-HBsAg (HBV carrier) status, or positive HCV status requires emergency treatment, use the following precautions**

- Consult the patient's physician regarding status
- If bleeding is likely during or after treatment, measure prothrombin time (PT) and bleeding time.
- All personnel in clinical contact with the patient should use full barrier technique, including masks, gloves, glasses or eye shields, and disposable gowns

- Aseptic techniques should be followed at all times. Minimize aerosol production by not using ultrasonic instrumentation, air syringe, or high-speed hand pieces.
- Remember that saliva contains a distillate of the virus. Pre-rinsing with chlorhexidine gluconate for 30 s is highly recommended

How to prevent

Sterilization and disinfection of dental equipment and instruments

Body fluids is also important since HCV can survive at room temperature on surfaces for more than five days, and HBV can survive for at least one week.

The CDC recommends cleaning exposed surfaces with a 1:10 dilution of bleach to water.

- All working surfaces and environmental surfaces should be wiped with 2% activated glutaraldehyde **(Cidex).**

Precautions

- All disposable items (e.g., gauze, floss, saliva ejectors, masks, gowns, gloves) should be placed in **a lined wastebasket**.
- After treatment, these items and all disposable covers should be bagged, **labeled,** and disposed of, following proper guidelines for bio-hazardous waste
- When the procedure is complete, all equipments **should be scrubbed and sterilized**. If an item cannot be sterilized or disposed of, it should not be used

Post-Exposure Prophylaxis

Step one: Treatment of the site of exposure

The site of exposure to infectious fluids should be washed as soon as possible using soap and water only,

- The exposed mucus membranes should be washed with water only.
- Eyes should be flushed with water and saline solution (if there was contact with potentially infectious fluids).
- Do not use caustics and do not rinse the wound with antiseptics and disinfectants.

Step two: Report and documentation

- Occupational exposure should be reported immediately. Details regarding the circumstances under which the exposure happened and the employee was administered prophylaxis should be recorded in the employee's medical record.

Post-Exposure Prophylaxis

Step three: Evaluation of sources

- Test the patient for anti-HBsAg, HCV and HIV antibodies
- Evaluation of "viral load" (the level of infectious particles in the blood)

Test the patient by rapid HIV test

Step four : Specific prophylaxis

- Any blood or body fluid exposure to an unvaccinated person **should lead to the initiation of the hepatitis B vaccine series** (at 0, 1, and 6 months). When **hepatitis B Immune Globulin** (HBIG) is indicated, it should be administered **as soon as possible** after the exposure (preferable **within 24 h**,
- Hepatitis B vaccine can be administered simultaneously with HBIG but at a separate site.
- **During the testing period**, Exposed person should abstain from changes in sexual activity, pregnancy, breastfeeding, or professional activities not to donate blood, plasma,
- **Testing for anti-HBs antibodies 1-2 months after the last dose of vaccine**

Dental Considerations and Management of Sickle Cell Patients

- For sickle cell patients, the most preferred safe analgesic treatment is a combination of paracetamol and codeine. In addition, morphine and its derivative analgesics can be used Aspirin should not be used to avoid the risk of acidosis.
- Extreme care should be taken regarding infections. Any source of infection either dental or non-dental source should be treated vigorously with SCD patients to decrease the chance of crisis caused by infections.
- **prophylactic antibiotics should be prescribed for them for prevention of infection**, if they will undergo special dental operation
- The pulp therapy is a case selection and it is not always a good option especially in the case of the possibility of treatment failure. At that point, extraction is the best option to eradicate the chances of failure. The latter treatment option should be exerted with extreme care to preserve as well as protect the fragile, osteoporotic bone from fracture. In addition, to decrease the risk of future osteonecrosis that may accompany traumatic extraction


قال رسول الله صلى الله عليه وسلم
مَنْ خَرَجَ فِي طَلَبِ العِلْمِ
فَهُوَ فِي سَبِيلِ اللهِ حَتَّى يَرْجِعَ
جامع الترمذي

Case Discussion

Dental Student With Active Replication

24-year-old student who has recently been admitted to dental school. He is hepatitis B surface antigen (HBsAg) positive, hepatitis B e antigen (HBeAg) positive, and anti-HBe negative. His HBV DNA is 80 million international units/mL and serum alanine transaminase (ALT) is 45 U/L. The Dean of the dental school has denied his matriculation pending your advice.

The CDC recommends that chronic HBV infection in itself not preclude the practice or study of medicine, surgery, dentistry, or allied health professions provided that the following criteria are met:

Case Discussion

Health care providers and students who perform invasive exposure-prone procedures should have oversight from an expert panel. The expert panel should include one or more persons with expertise in the provider's specialty, infectious diseases and hospital epidemiology, liver disease, the infected provider's occupational health, student health or primary care clinicians, ethicists, human resource professionals, hospital or school administrators, and legal counsel.

HBV-infected providers can conduct exposure-prone **procedures if a low or undetectable HBV viral load** (spontaneous or while on treatment) is documented by regular testing at least every six months.

A threshold of **1000** international units/mL is appropriate.

Spontaneous fluctuations **above this level** may occur and should prompt the HBV-infected provider to **abstain from performing exposure-prone procedures** while awaiting subsequent retesting, changing drug therapy, or other reasonable steps.

Case Discussion

Standard precautions should be rigorously adhered to in all settings.

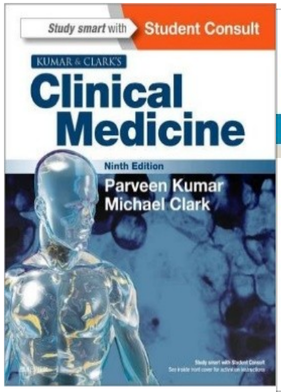
- Based on the above, **you can advise this patient to reconsider his plans to enter dental school**. Because he has high HBV DNA and low ALT levels, the likelihood of him responding (HBeAg loss) to current treatment is low and the chance of him clearing HBsAg is even more remote. He will likely need long-term if not life-long treatment and be subjected to frequent monitoring throughout his career

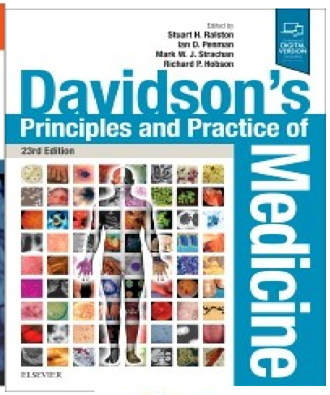
- The impact of chronic HBV infection with high level viremia on career paths of dental students is different from medical students who can choose career paths that do not involve performance of invasive procedures.

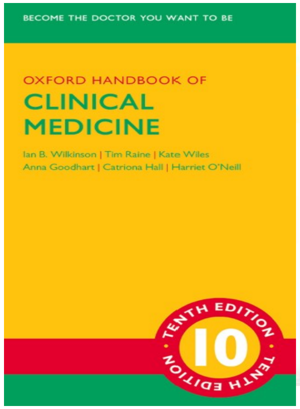
- However, if this young man is passionate about being a dentist, he should be allowed to do so by starting antiviral to keep serum HBV DNA suppressed and having regular monitoring to confirm that HBV DNA is below 1000 international units/mL.

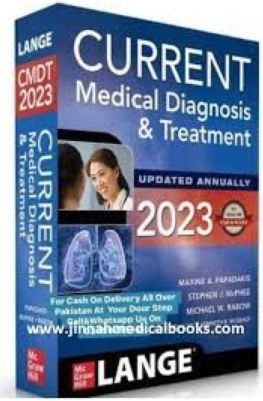
References & Further Readings

- **Kumar and Clark's Clinical Medicine,** 10th Edition
- **Davidson's Principles and Practice of Medicine**, 23rd Edition
- **Oxford Handbook of Clinical Medicine,** 10th Edition
- **CURRENT Medical Diagnosis and Treatment 2023** https://erj.ersjournals.com/content/61/4/2300239









THANK YOU